

Development of a Closed-Loop Feedback Control System for EXOHSPEC

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ABSTRACT

Optical stability in high-resolution spectrographs is strongly affected by variations in the ambient environment, particularly when the instrument operates in air rather than under vacuum. In EXOhSPEC, fluctuations in temperature, pressure and humidity alter the refractive index and induce measurable changes in both optical path length (OPL) and the position of the spectrum on the detector, particularly along the dispersion axis. To mitigate these effects, we implemented a real-time feedback strategy that measures OPL using a laser interferometer, tracks spectral motion using the guider camera, and applies discrete temperature corrections using a thermo-electric cooler (TEC). A hybrid model was developed that combines interferometric OPL residuals with pixel-based regression to estimate the temperature adjustment required to stabilise the optical response of the instrument. Across multiple runs, feedback reduced long-term OPL drift by a factor of ~ 2 – 5 and suppressed dispersion-axis motion to the level of a few 10^{-2} pixels min^{-1} using temperature steps of only 10–100 mK. The total actuation effort remained below 0.3 °C over hour-long timescales, indicating that control was achieved within the linear response regime of the enclosure. These results show that environmental coupling in a spectrograph can be measured and actively compensated in real time, without the need for vacuum isolation. The method establishes a framework for model-based control in ambient-air spectrographs and provides a foundation for extending stabilisation to additional actuators and wavelength-dependent control strategies.

Key words: instrumentation: spectrographs – techniques: radial velocities – methods: data analysis – techniques: image processing – atmospheric effects

1 INTRODUCTION

High-resolution radial-velocity (RV) spectrographs must maintain a stable spectral position on the detector in order to achieve sub- 1 m s^{-1} velocity precision. A displacement of ~ 0.01 pixels is equivalent to the Doppler signal of an Earth-like planet (Fischer et al. 2016; Halverson et al. 2016), meaning that even small changes in the optical environment can mimic a planetary signature. These shifts commonly arise from fluctuations in temperature, pressure and humidity inside the instrument, which modify the refractive index of air and consequently alter the optical path length (OPL) and the detected position of the spectrum.

State-of-the-art spectrographs such as HARPS, NEID and ESPRESSO therefore operate in vacuum-sealed and thermally regulated environments (Mayor et al. 2003; Robertson et al. 2019; Pepe et al. 2021). These passive isolation strategies successfully suppress refractive index variability, but incur high cost, extended equilibration times and complex infrastructure requirements. As a result, rapid prototyping, portability and iterative development are difficult to implement in conventional vacuum-based architectures.

An alternative approach is to accept that environmental variations will occur and instead *measure and compensate for their effects in real time*. This defines a closed-loop feedback strategy in which environmental fluctuations are sensed and actively corrected rather than eliminated. Similar principles have been demonstrated in laboratory

spectrographs using adaptive alignment and emission-line tracking (Jones et al. 2021; Petersburg et al. 2018), indicating that active control may provide a viable route to RV stability in ambient air.

EXOhSPEC is designed to investigate this possibility in a controlled optical system. Instead of regulating temperature alone, the goal is to stabilise the *position of the spectrum on the detector* by tracking the measured coupling between OPL variation, environmental parameters and image motion. If these relationships can be modelled accurately, a temperature correction applied via a thermo-electric cooler (TEC) may be sufficient to counteract environmental drift while operating outside a vacuum enclosure.

In this paper, we present a closed-loop hybrid feedback architecture for EXOhSPEC that combines real-time OPL monitoring, guider-centroid tracking and environmental sensing into a unified control law. The system is experimentally validated in Section 4, demonstrating sub-pixel stability in ambient air and establishing a framework for feedback-driven spectrograph control.

2 SYSTEM OVERVIEW

The EXOhSPEC spectrograph is designed to investigate whether high-precision spectral stability can be achieved in ambient air by actively correcting, rather than passively isolating, environmentally driven drift. The system does not attempt to hold a fixed temperature; instead, its objective is to stabilise the *position of the spectrum on the detector* by monitoring the optical response of the spectro-

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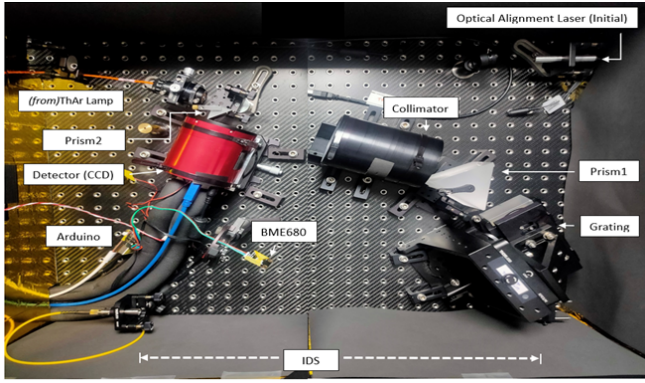


Figure 1. Internal structure of the modified EXOhSPEC, showing integration of the interferometric displacement sensor (IDS3010) and BME680 environmental sensor within the thermal enclosure box, mounted on a carbon fibre breadboard.

graph in real time and applying targeted temperature corrections using a thermo-electric cooler (TEC). The TEC acts as the sole actuator within the system, enabling temperature adjustments in discrete steps (typically 10–100 mK) with a thermal response time of 15–30 minutes.

Three observables are recorded continuously: the optical path length (OPL), the environmental parameters (temperature T , pressure P and humidity H), and the detector position of the spectrum. The OPL is measured using a laser interferometer (Attocube IDS3010), which provides sub-nanometre sensitivity along a fixed optical axis. Environmental sensors positioned near the dispersive optics record T , P and H , allowing the refractive index of air to be estimated via a modified Edlén formulation,

$$\eta = 1 + 7.86 \times 10^{-4} \frac{P}{273.15 + T} - 1.5 \times 10^{-11} RH (T^2 + 160), \quad (1)$$

where T is in $^{\circ}\text{C}$, P in hPa and RH is relative humidity (%). This treatment follows the wavelength-calibration guidelines provided by NIST,¹ offering a physically motivated way to link atmospheric variability to measurable shifts in OPL. Although the laboratory temperature is stabilised by an air-conditioning system at 22°C , slow residual drifts remain at the millikelvin level, and these form the basis of the feedback strategy discussed in Section 3.

The position of the spectrum on the detector is tracked using MaxIm DL in combination with a Python-based interface. Each exposure is acquired at a fixed cadence, and the CalcInformation() function returns the centroid coordinates of the spectral reference via a predefined aperture. This yields an (X, Y) position at every time step, from which the displacement is computed as

$$\Delta X(t) = X(t) - X_{\text{ref}}, \quad \Delta Y(t) = Y(t) - Y_{\text{ref}},$$

where $(X_{\text{ref}}, Y_{\text{ref}})$ is taken from a stable reference interval. Across all runs, the dispersion direction (ΔY) exhibited the strongest environmental coupling and was approximated via

$$\Delta Y(t) \approx \beta_0 + \beta_T \Delta T(t) + \beta_P \Delta P(t) + \beta_H \Delta H(t), \quad (2)$$

where β_T , β_P and β_H quantify the sensitivity of spectral position to changes in environmental conditions. This regression model forms the basis for estimating the temperature correction required to minimise both OPL residuals and pixel drift.

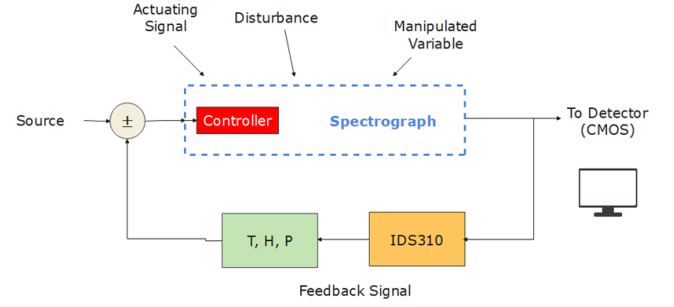


Figure 2. Flowchart of the singular control system. The process begins with data collection initiated through a web interface, utilizing various light sources. Measurements from the Interferometer and BME680 sensors are processed and integrated before transmission to the Control Unit. Spectrograph calibration is periodically evaluated, ensuring data acquisition proceeds only when calibration is successful.

All data are synchronised using a Python-based acquisition routine and logged at a one-minute cadence. Each feedback cycle consists of four stages: data acquisition, residual estimation against a rolling baseline, temperature correction proposal, and TEC actuation followed by a cooldown interval. This establishes a feedback loop driven by the *optical response of the instrument* rather than by explicit thermal control. In other words, the temperature is treated as the actuator rather than the quantity to be held constant. This distinction is central to the methodology and motivates the hybrid OPL–pixel formulation introduced in Section 3.

3 FEEDBACK MODELLING METHODOLOGY

The feedback system computes the temperature correction required to maintain optical stability in EXOhSPEC while operating in ambient air. Passive environmental control suppresses short-term fluctuations but does not compensate for slow drifts in optical response. Therefore, the optical state of the spectrograph must be *quantified*, expressed as a control variable, and converted into an actuation signal for the thermo-electric cooler (TEC). For this purpose, both OPL and detector-based spectral motion are treated as measurable proxies of optical stability. Three components form the basis of the feedback strategy: an OPL drift model, a pixel–temperature relationship, and a fusion method that yields a single temperature command for actuator control.

To verify that variations in OPL encode environmental changes, an initial calibration experiment was conducted without feedback. The interferometric OPL response was measured simultaneously with temperature, pressure and humidity under a controlled warm-up phase. Figure 3 shows that OPL follows environmental trends with clear temporal correlation, demonstrating that OPL can serve as a physically grounded proxy for thermo-optic effects and density fluctuations in ambient air. This observation motivates the use of OPL in the drift model of Section 3.1.

3.1 Optical-path drift model

The measured OPL deviation relative to a reference condition is defined as

$$y(t) = \Delta \text{OPL}(t), \quad (3)$$

¹ <https://emtoolbox.nist.gov/wavelength/documentation.asp>

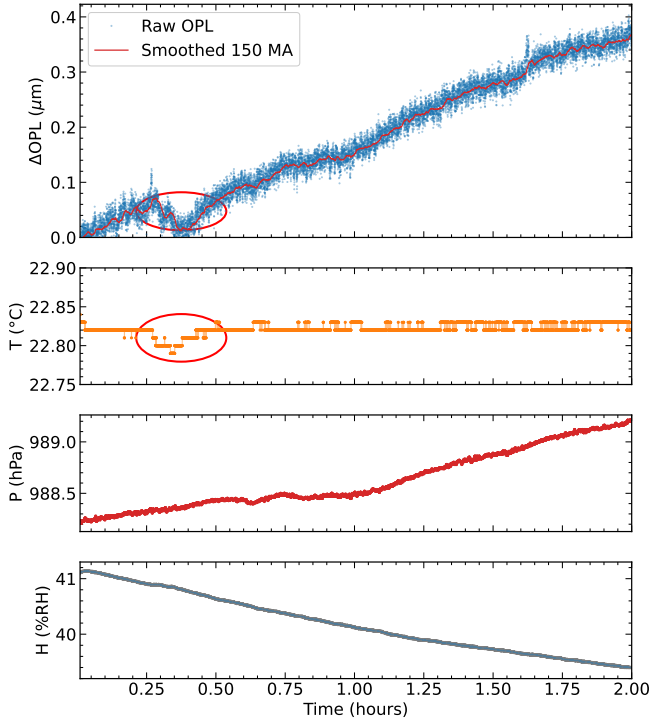


Figure 3. Open-loop calibration demonstrating that optical path length (OPL) carries environmental signatures. During a 90 min warm-up interval, the OPL variation correlates with pressure ($r \approx +0.90$) and humidity ($r \approx -0.88$), despite only minor changes in temperature ($< 0.05^{\circ}\text{C}$). A short temperature dip (circled) produces a $\sim 1.2 \mu\text{m}$ response in OPL, confirming its sensitivity to thermo-optic effects. This establishes OPL as a metrology channel from which g_T , g_P and g_H can be inferred.

where t is experiment time and $y(t)$ represents the optical path variation induced by environmental changes. The response is modelled as a combination of a slow baseline and the effects of temperature, pressure, and humidity:

$$\hat{y}(t) = a_0 + a_1 t + \sum_{k \in K} w_k (u_k * g_{\tau_k, \theta_k})(t), \quad (4)$$

where $u_k(t)$ are the input channels (T , P , and H), w_k are the corresponding weights, and $*$ denotes convolution with a first-order kernel,

$$g_{\tau, \theta}(t) = \frac{1}{\tau} \exp\left[-\frac{(t - \theta)}{\tau}\right] \mathbf{1}_{t \geq \theta}. \quad (5)$$

Here, τ represents the thermal response timescale of the enclosure and θ the delay before the environmental change is expressed in the OPL. This formulation provides a physically motivated way to separate deterministic drift from residual motion. From this model, the steady-state sensitivities of the OPL to temperature, pressure and humidity are quantified through the gains

$$g_T \approx 1.13 \mu\text{m K}^{-1}, \quad g_P \approx 0.82 \mu\text{m hPa}^{-1}, \quad g_H \approx -1.70 \mu\text{m } \%^{-1}. \quad (6)$$

None of these coefficients are assumed a priori. They are obtained directly from the OPL time series by isolating the independent contributions of temperature, pressure and humidity during calibration runs without feedback. In particular, g_T is derived by fitting the OPL response to temperature after subtracting pressure and humidity terms, while g_P and g_H are extracted in the same manner from their respective OPL correlations. These fitted values are subsequently used as

seed coefficients for feedback runs. In the real-time implementation, the model structure is kept fixed but the gains are re-estimated during the warm-up phase, allowing g_T , g_P and g_H to adapt to the prevailing conditions while the optical configuration remains unchanged.

3.2 OPL residual and equivalent temperature error

The residual OPL signal is obtained by removing the baseline trend and pressure/humidity components:

$$\text{OPL}_{\text{resid}}(t) = \text{OPL}(t) - (a_0 + a_1 t) - g_P \Delta P(t) - g_H \Delta H(t). \quad (7)$$

Since g_T quantifies the temperature sensitivity of OPL, the residual can be interpreted as the *equivalent temperature error*,

$$e_T^{\text{OPL}}(t) = -\frac{\text{OPL}_{\text{resid}}(t)}{g_T}, \quad (8)$$

which specifies the temperature change required to null the observed optical-path deviation.

3.3 Pixel-based regression model

Spectral motion on the detector provides an independent measure of optical stability. For each frame, the centroid of the spectral reference is extracted in (X, Y) coordinates. Over thermal timescales, this motion is approximated as

$$\begin{bmatrix} \Delta X(t) \\ \Delta Y(t) \end{bmatrix} \approx \beta \begin{bmatrix} \Delta T(t) \\ \Delta P(t) \\ \Delta H(t) \end{bmatrix} + \varepsilon(t), \quad (9)$$

where β is the sensitivity matrix and $\varepsilon(t)$ represents residual and noise components. The dispersion axis (ΔY) exhibits the strongest and most consistent environmental response. This is expected, since refractive index variations predominantly affect the optical path in the beam propagation direction, which maps directly onto the wavelength (dispersion) axis of the detector. As a result, OPL fluctuations are more likely to manifest as shifts in ΔY , whereas ΔX tends to remain comparatively less sensitive and is often influenced by minor alignment effects. Accordingly, a rolling linear regression is applied to extract $\beta_{Y,T}$, and once the pressure and humidity contributions are removed, the equivalent temperature error is defined as

$$e_T^{\text{PX}}(t) = -\frac{\Delta Y(t) - \beta_{Y,P} \Delta P(t) - \beta_{Y,H} \Delta H(t)}{\beta_{Y,T}}. \quad (10)$$

This represents the temperature adjustment required to minimise the spectral drift on the detector.

3.4 Hybrid fusion of OPL and pixel channels

The OPL and pixel channels probe different aspects of the system: interferometric data offer a low-noise, physically grounded measurement, while pixel motion directly relates to spectral stability. To use both sources of information, the equivalent temperature estimates are fused as

$$e_T(t) = w e_T^{\text{OPL}}(t) + (1 - w) e_T^{\text{PX}}(t), \quad (11)$$

with $0.8 \lesssim w \lesssim 0.95$ during typical runs. The fused signal is passed to a proportional-integral controller:

$$u_{\text{PI}}(t) = K_p e_T(t) + K_i \int_0^t e_T(\tau) d\tau, \quad (12)$$

and the corresponding TEC command is

$$\Delta T_{\text{cmd}}(t) = \text{clip}\left[u_{\text{PI}}(t), \Delta T_{\text{min}}, \Delta T_{\text{max}}\right], \quad (13)$$

where ΔT_{min} and ΔT_{max} define conservative bounds for thermal actuation. Each correction is followed by a cooldown period to respect the thermal response time of the enclosure.

This modelling strategy enables real-time stabilisation based on the *optical response* of the spectrograph, providing the foundation for the experimental validation presented in Section 4.

4 EXPERIMENTAL PROTOCOL AND RESULTS

All experiments were conducted with EXOhSPEC operating in ambient laboratory conditions, with temperature regulated solely by the thermo-electric cooler (TEC) introduced in Section 2. Each run followed a two-stage approach: an initial interval under fixed TEC setpoint (referred to as the warm-up phase), followed by a feedback phase in which the OPL@PX controller was activated. This structure allowed the baseline OPL drift, environmental sensitivities, and thermal response time to be characterised before closed-loop actuation began.

Data acquisition: All channels were synchronised at a one-minute cadence using a Python-based logging routine. The data included: TEC setpoint and internal temperature; OPL from the IDS3010 interferometer; environmental parameters (T, P, H) from ECU/BME sensors; and centroid positions (X, Y) extracted from MaxIm DL. This cadence captured the slow thermal evolution of the enclosure while avoiding artefacts from rapid sensor cycling or local heating.

Warm-up and model calibration: A linear trend was removed from the OPL time series to define its baseline drift. The sensitivities (g_T, g_P, g_H) were determined following the method of Section 3.1, while the dominant pixel-temperature coefficient $\beta_{Y,T}$ was extracted via a rolling regression of ΔY against the environmental inputs, as described in Section 3.3. These parameters varied slightly across experiments but remained within experimental uncertainty and were used directly to initialise the feedback phase.

Feedback phase: Once calibration was complete, the fused temperature error $e_T(t)$ was computed according to Section 3.4 and passed to the PI controller. Temperature corrections were applied as discrete steps, generally between 10–100 mK, followed by cooldown intervals of 15–30 minutes to respect the enclosure’s thermal response time. The cumulative TEC actuation remained below 0.3°C in all runs, and no retuning of gains was required between experiments. The slow actuation strategy ensured that feedback operated entirely within a quasi-linear regime.

Time-series behaviour: Figure ?? shows a representative experiment processed using this protocol. During warm-up, both the OPL residual and dispersion-axis drift increased steadily due to environmental variability. Once feedback was enabled, the residual OPL rapidly flattened and the drift in ΔY was reduced. Temperature actuation corresponded directly to changes in slope, and no oscillatory behaviour was observed. The concurrent evolution of OPL, centroid motion, pressure and humidity indicates that the controller was responding to changes in air-density rather than intrinsic spectrograph instabilities.

Run-to-run comparison: To assess repeatability, all experiments were plotted on a common time axis. Early runs (Exp-A to Exp-C) showed monotonic drift exceeding 10 px hr⁻¹ on the disper-

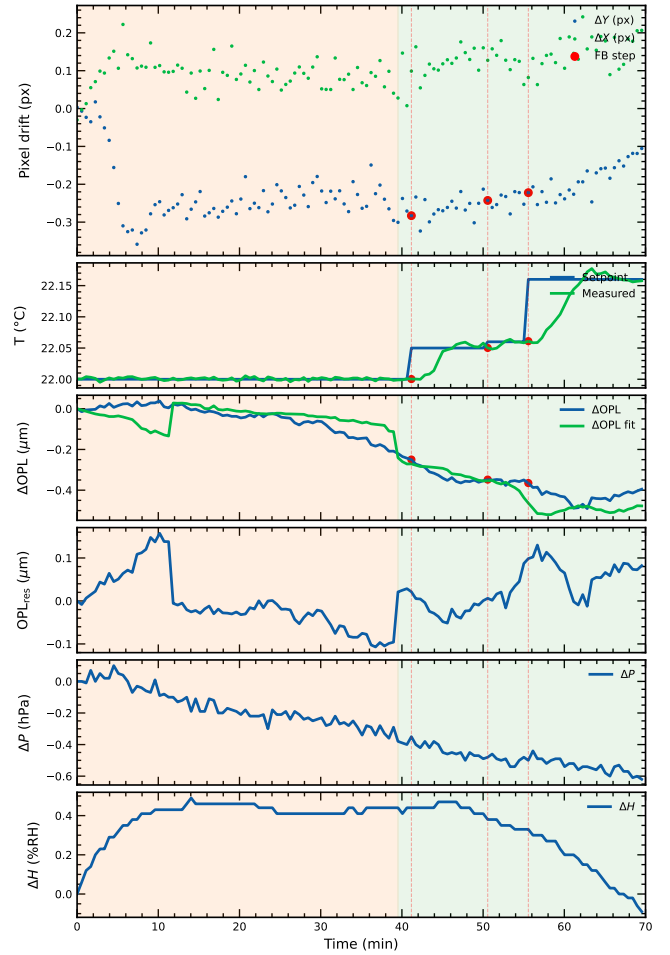


Figure 4. Multi-channel time series from the Hybrid 4 feedback experiment (Exp-Ea). The warm-up interval is followed by activation of the hybrid controller, after which the dispersion-axis drift remains within $\sigma(\Delta Y) \approx 0.04$ px and the cross-dispersion axis remains near zero with no systematic trend. The OPL residual shows reduced slope and scatter during feedback, while pressure and humidity exhibit slow monotonic changes that correlate with the remaining pixel drift. Feedback steps are indicated by dashed vertical lines and correspond to small, discrete temperature adjustments ($|\Delta T| < 50$ mK) without inducing thermal overshoot.

sion axis, whereas the hybrid controller suppressed this drift to the 10⁻² px min⁻¹ level from Exp-D onward. The full centroid trends are shown in Figure 5, and the associated drift rates are reported in Table 1. Experiment F provided the lowest overall drift and represents the current best-performing result.

Environmental coupling: In the most stable run, the drift in ΔY exhibited strong correlation with pressure and humidity:

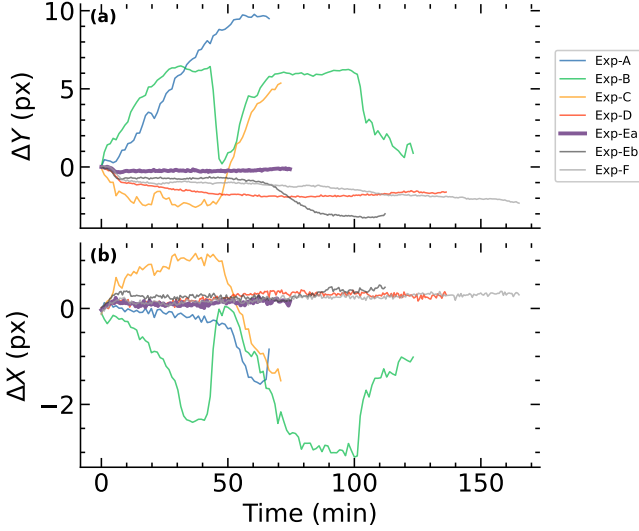
$$r(\Delta Y, P) \approx -0.93, \quad r(\Delta Y, H) \approx +0.95,$$

indicating that fluctuations in atmospheric density can influence spectral position even under temperature control. Although the feedback suppressed the cumulative drift, the residual behaviour suggests a secondary environmental actuator (humidity or pressure regulation) may be beneficial in future versions of the controller.

Overall, the results confirm that the OPL@PX strategy consistently reduces both OPL residual drift and dispersion-axis motion, delivering sub-pixel-per-hour stability using sparse temperature adjust-

Table 1. Measured centroid drift rates across experiments (warm-up + feedback).

Experiment	slope ΔY (px/min)	slope ΔY (px/hr)	slope ΔX (px/min)	slope ΔX (px/hr)
Exp-A	+0.1691	10.1483	-0.0212	-1.2737
Exp-B	-0.0032	-0.1891	-0.0140	-0.8399
Exp-C	+0.0911	5.4636	-0.0224	-1.3453
Exp-D	-0.0062	-0.3708	+0.0016	0.0952
Exp-Ea	+0.0001	0.0062	+0.0011	0.0653
Exp-Eb	-0.0056	-0.3379	-0.0000	-0.0030
Exp-F	-0.0107	-0.6420	+0.0013	+0.0772

**Figure 5.** (a) Dispersion-axis centroid drift ΔY and (b) cross-dispersion drift ΔX across all experiments (Exp-A to Exp-F). The time axes are aligned to enable direct comparison of controller performance. Experiments Exp-Ea/Eb exhibit the flattest trajectories, indicating near-sub-pixel stability and successful suppression of systematic drift. The plot highlights the progressive improvement achieved through the hybrid OPL@PX feedback formulation.

ments. The final experiment demonstrates that real-time inference of environmental sensitivity is possible in ambient conditions and that EXOHSPEC can be stabilised without vacuum isolation when the environmental coupling is quantified and integrated into the control law.

5 DISCUSSION AND CONCLUSIONS

The experiments presented in this work demonstrate that spectral stability in EXOHSPEC can be achieved in ambient air through model-based feedback rather than passive thermal isolation. Across all runs, the hybrid OPL@PX controller consistently reduced long-term drift in both optical path length and detector position. In the most stable configuration (Exp-F), the dispersion-axis drift was constrained to the order of a few 10^{-2} pixels min^{-1} , while the total TEC actuation remained below 0.3°C . These observations indicate that EXOHSPEC responds predictably to thermal perturbations and that closed-loop control can maintain an approximately steady operating point without aggressive actuation.

The achieved stability is within the regime required for precise spectral extraction when combined with an accurate wavelength ref-

erence. Although vacuum enclosures remain a proven mitigation strategy, they impose complexity, long equilibration times and limited flexibility during instrument development. The results obtained here suggest an alternative approach: rather than eliminating air, its optical influence is measured and actively compensated in real time. This may be particularly advantageous for laboratory prototypes, validation test-beds and transportable spectrographs, where rapid reconfiguration is needed and environmental isolation is impractical.

A key outcome of these experiments is the demonstration that the dominant environmental coupling can be inferred in situ using OPL residuals and image motion. The model formulation uses measurable physical quantities and requires no empirical offset terms, allowing parameters such as g_T and $\beta_{Y,T}$ to remain transferable across runs. However, several experiments showed persistent correlations between pixel drift and pressure/humidity, suggesting that a purely thermal actuator may not be sufficient under rapidly changing environmental conditions. This motivates future exploration of secondary actuators (e.g. localised heaters or opto-mechanical elements) or multi-input controller logic.

Further refinements may include improved centroiding accuracy, higher temporal sampling in the warm-up phase and extended calibration of the refractive index model. Adaptive variants of the controller, incorporating rolling model identification, could reduce latency and operate more effectively during diurnal transitions or HVAC shutdown. The methodology developed here can also be applied to stabilise auxiliary subsystems, including wavelength references or dispersive optics, where slow alignment changes may accumulate over time.

Overall, the EXOHSPEC test-bed demonstrates that environmental coupling in a spectrograph can be treated as a measurable quantity rather than solely a noise source. By combining interferometric sensing, image tracking and physically motivated feedback, fine-scale stabilisation was maintained in ambient air without vacuum isolation. This work therefore occupies an intermediate region between passive insulation and post-processing, and establishes a framework for future hybrid architectures that integrate metrology, modelling and actuation to preserve instrumental stability on scientific timescales.

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DATA AVAILABILITY

Data underlying this article will be shared on reasonable request to the corresponding author.

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